

TSIA2 Math Practice Test 2

PART A — QUESTIONS

TSIA2-M2-001

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A student buys 3 notebooks for \$2.75 each and 2 folders for \$1.40 each. If the student pays with a \$20 bill, how much change should the student receive?

- A) \$8.95
- B) \$9.15
- C) \$10.25
- D) \$11.75

TSIA2-M2-002

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A line has a slope of 3 and passes through the point (2, 10). Which equation represents the line?

- A) $y = 3x + 4$
- B) $y = 3x + 8$
- C) $y = 2x + 3$
- D) $y = 10x + 3$

TSIA2-M2-003

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A triangular banner has a base of 14 inches and a height of 9 inches. What is the area of the banner?

- A) 23 square inches
- B) 46 square inches
- C) 63 square inches
- D) 126 square inches

TSIA2-M2-004

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The table shows the number of books checked out from a library during one week.

Monday: 42

Tuesday: 38

Wednesday: 51

Thursday: 49

Friday: 60

What is the range of the data?

- A) 18
- B) 20
- C) 22
- D) 24

TSIA2-M2-005

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A community college parking pass costs \$135 for a 15-week semester. What is the cost per week?

- A) \$7
- B) \$8
- C) \$9
- D) \$10

TSIA2-M2-006

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Simplify the expression $4(3x - 2) - 5(x + 6)$.

- A) $7x - 38$
- B) $7x + 22$
- C) $17x - 38$
- D) $17x + 22$

TSIA2-M2-007

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A circular garden has a diameter of 18 feet. What is its circumference?

- A) 9π feet
- B) 18π feet
- C) 36π feet
- D) 81π feet

TSIA2-M2-008

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A class has 12 students who prefer online textbooks, 18 students who prefer printed textbooks, and 10 students who have no preference. If one student is chosen at random, what is the probability that the student prefers printed textbooks?

- A) $3/10$
- B) $2/5$
- C) $9/20$
- D) $1/2$

TSIA2-M2-009

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A science lab sample has a mass of 72 grams. After drying, the mass is 54 grams. By what percent did the mass decrease?

- A) 18%
- B) 20%
- C) 25%
- D) 33%

TSIA2-M2-010

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A ride-share company charges a base fee of \$4 plus \$1.75 per mile. If m represents the number of miles and C represents the total cost in dollars, which equation models this situation?

- A) $C = 4m + 1.75$
- B) $C = 1.75m + 4$
- C) $C = 5.75m$
- D) $C = 4(m + 1.75)$

TSIA2-M2-011

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular classroom floor is 28 feet long and 22 feet wide. Carpet costs \$3.50 per square foot. What is the total cost to carpet the floor?

- A) \$175
- B) \$616
- C) \$1,750
- D) \$2,156

TSIA2-M2-012

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The scores on a quiz were 68, 72, 75, 81, 84, 90, and 91. What is the median score?

- A) 75
- B) 81
- C) 84
- D) 90

TSIA2-M2-013

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A printer produces 36 pages in 4 minutes. At this same rate, how long will it take the printer to produce 99 pages?

- A) 9 minutes
- B) 10 minutes
- C) 11 minutes
- D) 12 minutes

TSIA2-M2-014

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

For the function $g(x) = x^2 - 4x$, what is $g(6)$?

- A) 8
- B) 12
- C) 24
- D) 60

TSIA2-M2-015

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A square patio has an area of 196 square feet. What is the perimeter of the patio?

- A) 14 feet
- B) 28 feet
- C) 56 feet
- D) 784 feet

TSIA2-M2-016

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A box contains 6 granola bars: 2 chocolate, 3 peanut butter, and 1 oatmeal. If one bar is selected at random and then another bar is selected without replacement, what is the probability that the first bar is chocolate and the second bar is oatmeal?

- A) $\frac{1}{18}$
- B) $\frac{1}{15}$
- C) $\frac{1}{9}$
- D) $\frac{1}{6}$

TSIA2-M2-017

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A laptop is priced at \$840. A store offers a 15% student discount. What is the discounted price before tax?

- A) \$126

- B) \$714
- C) \$725
- D) \$825

TSIA2-M2-018

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve the system of equations:

$$2x + y = 17$$

$$x - y = 4$$

What is the value of x?

- A) 5
- B) 6
- C) 7
- D) 13

TSIA2-M2-019

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A ramp rises 5 feet over a horizontal distance of 12 feet. What is the length of the ramp?

- A) 13 feet
- B) 15 feet
- C) 17 feet
- D) 60 feet

TSIA2-M2-020

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The table shows the numbers of students enrolled in four workshop sections.

Section A: 18 students

Section B: 22 students

Section C: 25 students

Section D: 35 students

What percent of the students are enrolled in Section D?

- A) 25%
- B) 30%
- C) 35%
- D) 40%

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M2-001 — Correct: A — Correct Answer: \$8.95 — Explanation: The notebooks cost $3 \times \$2.75 = \8.25 , and the folders cost $2 \times \$1.40 = \2.80 . The total is \$11.05, so the change from \$20 is \$8.95.

TSIA2-M2-002 — Correct: A — Correct Answer: $y = 3x + 4$ — Explanation: Use $y = mx + b$ with slope 3. Substituting (2, 10) gives $10 = 3(2) + b$, so $b = 4$.

TSIA2-M2-003 — Correct: C — Correct Answer: 63 square inches — Explanation: The area of a triangle is $\frac{1}{2} \cdot \text{base} \cdot \text{height}$, so the area is $\frac{1}{2} \cdot 14 \cdot 9 = 63$ square inches.

TSIA2-M2-004 — Correct: C — Correct Answer: 22 — Explanation: The range is the greatest value minus the least value: $60 - 38 = 22$.

TSIA2-M2-005 — Correct: C — Correct Answer: \$9 — Explanation: Divide the total cost by the number of weeks: $\$135 \div 15 = \9 per week.

TSIA2-M2-006 — Correct: A — Correct Answer: $7x - 38$ — Explanation: Distribute to get $12x - 8 - 5x - 30$. Combining like terms gives $7x - 38$.

TSIA2-M2-007 — Correct: B — Correct Answer: 18π feet — Explanation: The circumference of a circle is $C = \pi d$. With diameter 18 feet, $C = 18\pi$ feet.

TSIA2-M2-008 — Correct: C — Correct Answer: $9/20$ — Explanation: There are $12 + 18 + 10 = 40$ students total. The probability of choosing a student who prefers printed textbooks is $18/40 = 9/20$.

TSIA2-M2-009 — Correct: C — Correct Answer: 25% — Explanation: The decrease is $72 - 54 = 18$ grams. The percent decrease is $18/72 = 0.25$, or 25%.

TSIA2-M2-010 — Correct: B — Correct Answer: $C = 1.75m + 4$ — Explanation: The total cost is the per-mile charge times the number of miles plus the base fee, so $C = 1.75m + 4$.

TSIA2-M2-011 — Correct: D — Correct Answer: \$2,156 — Explanation: The floor area is $28 \times 22 = 616$ square feet. At \$3.50 per square foot, the total cost is $616 \times 3.50 = \$2,156$.

TSIA2-M2-012 — Correct: B — Correct Answer: 81 — Explanation: The scores are already in order. With 7 values, the median is the 4th value, which is 81.

TSIA2-M2-013 — Correct: C — Correct Answer: 11 minutes — Explanation: The printer produces $36 \div 4 = 9$ pages per minute. To produce 99 pages, it needs $99 \div 9 = 11$ minutes.

TSIA2-M2-014 — Correct: B — Correct Answer: 12 — Explanation: Substitute 6 for x : $g(6) = 6^2 - 4(6) = 36 - 24 = 12$.

TSIA2-M2-015 — Correct: C — Correct Answer: 56 feet — Explanation: A square with area 196 has side length $\sqrt{196} = 14$ feet. Its perimeter is $4 \times 14 = 56$ feet.

TSIA2-M2-016 — Correct: B — Correct Answer: $1/15$ — Explanation: The probability of first selecting chocolate is $2/6$. After that, 5 bars remain, including 1 oatmeal bar. The probability is $2/6 \times 1/5 = 2/30 = 1/15$.

TSIA2-M2-017 — Correct: B — Correct Answer: \$714 — Explanation: The discount is 15% of \$840, or $0.15 \times 840 = \$126$. The discounted price is $\$840 - \$126 = \$714$.

TSIA2-M2-018 — Correct: C — Correct Answer: 7 — Explanation: Add the equations to eliminate y : $(2x + y) + (x - y) = 17 + 4$, so $3x = 21$ and $x = 7$.

TSIA2-M2-019 — Correct: A — Correct Answer: 13 feet — Explanation: The ramp is the hypotenuse of a right triangle with legs 5 and 12. Its length is $\sqrt{5^2 + 12^2} = \sqrt{169} = 13$ feet.

TSIA2-M2-020 — Correct: C — Correct Answer: 35% — Explanation: The total number of students is $18 + 22 + 25 + 35 = 100$. Section D has 35 out of 100 students, which is 35%.